

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. Phase coexistence is represented by an exact microscopic labelled mixture of genuine phase-restricted laws. The physical rate is obtained by contracting the joint label/state LDP, not by conjugating a maximum pressure, and phase-local complex charts remain compatible with Lee–Yang pinching. Labelled projective passage, thin-shell conditioning, exact finite-centred likelihoods, and nonlinear values form a standalone commutation theorem.

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1. ROUND-NINE RECONSTRUCTION: EXACT PHASE LABELS, MICROSCOPIC MIXTURES, AND COMMUTING CONTRACTIONS

The phase decomposition is now part of the microscopic probability space. A minimum of phase rates is obtained by contracting an exact labelled mixture, not by taking the conjugate of a maximum pressure and not by discarding an $e^{-c\mu}$ remainder, which can change a large-deviation rate. Phase-local complex charts are used only for local response and likelihoods; coexistence is handled by the label variable and allows Lee–Yang pinching.

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1.1. Exact labelled finite-volume laws. Let $\mathfrak{A}$ be a compact finite or finite-dimensional phase-label space. For every ε let ρ_ε be a probability on $\mathfrak{A}$ and let $\mathbb{P}_{\varepsilon,\alpha}$ be a genuine normalized microscopic law on a common measurable state space E_ε . The labelled law is

$$\bar{\mathbb{P}}_\varepsilon(d\alpha, dx) = \rho_\varepsilon(d\alpha) \mathbb{P}_{\varepsilon,\alpha}(dx).$$

The physical unlabelled law is its exact projection. For Sinai phases the labels are recurrent components/face phases from A3; for hard spheres they are the B1/B2 prepared exposed phases and coexistence components. Restricted partition functions and boundary conditions defining each law are included in the platform theorem.

Assumption 1.1 (Subexponential phase prior). The priors satisfy an LDP at speed μ_ε with good rate $J(\alpha)$, and on every compact set of labels their masses are bounded below by $e^{-o(\mu_\varepsilon)}$ relative to the appropriate neighbourhoods. There is no discarded signed or total-variation remainder.

Theorem 1.2 (Joint labelled platform LDP). *Suppose the platform theorems give, uniformly on compact label sets, a good conditional LDP for X_ε under $\mathbb{P}_{\varepsilon,\alpha}$ with rate $I_\alpha(x)$, including boundary face recovery. Then (α, X_ε) under $\bar{\mathbb{P}}_\varepsilon$ satisfies the good joint LDP*

$$\bar{I}(\alpha, x) = J(\alpha) + I_\alpha(x).$$

Consequently the exact physical mixture satisfies

$$I(x) = \inf_{\alpha \in \mathfrak{A}} \{J(\alpha) + I_\alpha(x)\}.$$

For a finite zero-cost prior this is $\min_\alpha I_\alpha(x)$ because of the microscopic mixture theorem, not because $(\max Q_\alpha)^* = \min Q_\alpha^*$.

Proof. For a closed $F \subset \mathfrak{A} \times E$, cover its compact α -projection by finitely many uniform conditional-LDP charts and combine the prior upper bound with the conditional upper bounds. Exponential tightness makes the rate good. For an open neighbourhood of (α, x) , use the prior lower bound and the phase-specific microscopic recovery supplied by A3 or B2/B1, including its face recovery when x lies at the boundary of the phase mean set. This proves the joint lower bound. Contracting the exact label coordinate gives the physical rate. No exponentially small term is ignored: a phase with prior cost J appears explicitly in the infimum. $\square$

1.2. Phase-local complex response charts. For a fixed phase label α , let

$$Z_{\varepsilon,\alpha}(\theta) = \mathbb{E}_{\varepsilon,\alpha} e^{\mu_\varepsilon \langle \theta, X_\varepsilon \rangle}.$$

On a regular phase chart the A2 or B2 transfer/cluster theorem gives a zero-free complex neighbourhood of a compact real source set and an analytic pressure Q_α . No one such neighbourhood is asserted across a coexistence locus. The physical mixture partition function is the positive sum

$$Z_\varepsilon(\theta) = \int Z_{\varepsilon,\alpha}(\theta) \rho_\varepsilon(d\alpha),$$

and may have complex zeros at distance $O(\mu_\varepsilon^{-1})$ from coexistence.

Theorem 1.3 (Local derivatives and exact finite-centred likelihood). *On a regular fixed-phase chart all pressure derivatives commute with the microscopic limit. For a finite-dimensional local direction h , the exact Radon–Nikodym likelihood within that phase is*

$$L_{\varepsilon,\alpha}(h) = \exp \left\{ \sqrt{\mu_\varepsilon} h \cdot (X_\varepsilon - DQ_{\varepsilon,\alpha}(\theta)) - \mu_\varepsilon \left[Q_{\varepsilon,\alpha}(\theta + h/\sqrt{\mu}) - Q_{\varepsilon,\alpha}(\theta) - \frac{1}{\sqrt{\mu}} DQ_{\varepsilon,\alpha}(\theta) h \right] \right\}.$$

It has expectation one exactly and converges jointly with the centred field to the Cameron–Martin likelihood. At coexistence, the phase label and its Bayesian weight remain in the local experiment.

Proof. Cauchy estimates on the phase-local zero-free chart give convergence of all finite derivatives. The displayed expression is the exact exponential tilt normalized by the finite pressure, hence is a mean-one likelihood. Taylor expansion at the exact finite mean and the A2/B3 Gaussian limit give the Cameron–Martin law. For several active labels, apply the statement conditionally and update the finite label weights by their exact likelihoods. No common analytic logarithm is used. $\square$

1.3. Faces and projective passage. Let π_m be an increasing family of finite continuous projections. The platform theorem provides for every label a full finite-dimensional LDP on the closed phase-attainable set $C_{m,\alpha}$. Interior points are recovered by finite tilts. Boundary faces are recovered by the actual A3 survivor/renewal construction or the B2/B1 constrained activity construction, not by merely sending an abstract normal source to infinity.

Theorem 1.4 (Labelled projective LDP and topology upgrade). *The joint labelled finite projections have compatible good rates*

$$\bar{I}_m(\alpha, y) = J(\alpha) + I_{m,\alpha}(y).$$

Dawson–Gärtner yields a projective joint LDP with rate

$$\bar{I}(\alpha, x) = \sup_m \bar{I}_m(\alpha, \pi_m x).$$

If the platform has exponential containment in a stronger weighted topology and every finite-rate point has a recovery sequence contained in one compact weighted sublevel, then the LDP upgrades to that topology. Contracting the label after this passage gives the physical rate of Theorem 1.2; no interchange of an arbitrary projective supremum with a phase minimum is used.

Proof. Compatibility follows from contraction of the exact joint finite-dimensional laws. Dawson–Gärtner gives the projective theorem. For a target-open set and a finite-rate point, choose the platform recovery inside a compact rate sublevel; on that compact the projective and weighted topologies agree, so a finite projective neighbourhood lies inside the target-open set. The recovery law has exponentially negligible mass outside the compact by the platform source tilt, providing the strong-topology lower

bound. Upper bounds use exponential containment. The label is retained until the final ordinary contraction. $\square$

1.4. Thin shells across phases. Let $C(X_\varepsilon)$ be a finite lattice–continuous constraint. For every active phase, A2 or B1 supplies a phase-specific exact two-saddle coefficient $\mathfrak{c}_{\varepsilon,\alpha}(H, a)$, uniform on the compact active label set.

Theorem 1.5 (Exact labelled thin-shell conditioning). *For a bounded source H , the conditioned physical pressure is obtained from the exact finite sum/integral*

$$\frac{\int \rho_\varepsilon(d\alpha) e^{\mu_\varepsilon q_{\varepsilon,\alpha}(H,a)} \mathfrak{c}_{\varepsilon,\alpha}(H, a)}{\int \rho_\varepsilon(d\alpha) e^{\mu_\varepsilon q_{\varepsilon,\alpha}(0,a)} \mathfrak{c}_{\varepsilon,\alpha}(0, a)}.$$

Laplace’s method on the compact label space gives the minimum constrained phase cost and retains the normalized coefficient weights among tied phases. Conditioning, typed contraction, local Gaussian tangent, and nonlinear dynamic programming commute after the label is included in the state.

Proof. Disintegrate the exact labelled law before conditioning and apply the phase-specific coefficient theorem. Uniformity allows Laplace’s method over α ; tied minimizers retain their prior and determinant weights rather than being merged into a nonexistent common saddle. Continuous contractions and dynamic values may then be applied to the joint labelled state. Local Gaussian tangents use Theorem 1.3 conditionally on the selected phase. $\square$

1.5. The standalone commutation theorem.

Theorem 1.6 (Exact phase-labelled commutation principle). *For every platform satisfying the explicitly proved A2/A3 or B1/B2/B3/B4 interfaces, the following diagram commutes on the exact labelled law:*

phase preparation $\longrightarrow$ path pressure/LDP $\longrightarrow$ conditioning and typed contraction $\longrightarrow$ Gaussian likelihood

The physical unlabelled statement is obtained by one final contraction of the phase label. The theorem permits phase coexistence and Lee–Yang pinching, never infers a minimum rate from a maximum pressure alone, and does not discard exponential remainders.

Proof. The joint LDP and physical rate are Theorem 1.2; local likelihood and Gaussian passage are Theorem 1.3; projective/topology passage is Theorem 1.4; and exact conditioning is Theorem 1.5. Each operation is performed before contracting the exact label, so the ordinary contraction and disintegration identities prove commutation. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine

dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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